

Most NBT tags are not kept when a mob converts to another mob

MC-88967 Entities

Status	Resolved / Fixed
Affected	Minecraft 15w38b, Minecraft 15w44b, Minecraft 1.9 Pre-Release 2, Minecraft 16w42a, Minecraft 1.12, Minecraft 1.12.2 (+62 more)
Fixed in	24w36a
Reported	2015-09-17

STEPS TO REPRODUCE

1. Launch Minecraft and create a Creative world.
2. Run `/summon minecraft:zombie_villager ~ ~ ~ {Health:10.0f,Invulnerable:1b,Fire:100s,Tags:["foo","bar"],ConversionTime:0}`.
3. Wait until the zombie villager converts into a villager.
4. Run `/data get entity @e[type=villager,limit=1,sort=nearest]`.
5. Observe that NBT tags such as `Health`, `Invulnerable`, `Fire`, and `Tags` are not preserved on the converted villager.

OBSERVED BEHAVIOUR

Actual behavior

When an entity converts, the original entity is replaced by a new entity of the target type. Only a limited set of data is copied to the new entity. Most other entity and mob NBT tags are discarded instead of being transferred.

For example, summoning a zombie villager with custom data:

```
``mcfunction
/summon minecraft:zombie_villager ~ ~ ~
{Health:10.0f,Invulnerable:1b,Fire:100s,Tags:["foo","bar"],ConversionTime:0}
...
```

After the conversion completes, querying the resulting villager:

```
``mcfunction
/data get entity @e[type=minecraft:villager,limit=1,sort=nearest]
...
```

shows that values such as `Health`, `Invulnerable`, `Fire`, and the entity `Tags` are not retained. The same loss occurs for the corresponding tags listed in the examples when other mobs convert, including equipment, attributes, effects, persistence, leash/passenger data, villager data, and other entity state.

`CustomName` and `NoAI` are preserved, demonstrating that conversion can transfer entity data, but there is no consistent preservation of the remaining applicable NBT. The issue also occurs when a mob splits into entities of the same type, such as a slime splitting into smaller slimes or a magma cube splitting into smaller magma cubes.

EXPECTED BEHAVIOUR

When a mob transforms into another mob, its applicable NBT data should be transferred to the resulting entity rather than reset or discarded. Properties such as health, invulnerability, fire state, custom tags, effects, equipment, persistence, and other supported entity data should remain consistent after conversion. This should also apply when mobs split into smaller entities, preserving the relevant data for the resulting mobs.

ENVIRONMENT

Minecraft Java Edition Versions: 15w38b, 15w44b, 1.9 Pre-Release 2, 16w42a, 1.12, 1.12.2, 18w02a, 18w11a, 1.13-pre4, 1.13-pre5, 1.13-pre6, 1.13, 1.13.1, 1.13.2, 19w02a, 19w12b, 19w13b, 1.14.4, 19w34a, 19w35a, 19w39a, 1.15.1, 1.15.2, 20w08a, 20w09a, 20w10a, 20w11a, 20w12a, 20w15a, 20w17a, 20w18a, 20w19a, 1.16 Pre-release 2, 1.16 Pre-release 3, 1.16 Pre-release 5, 1.16.1, 20w27a, 20w30a, 1.16.2, 1.16.3, 1.16.4, 20w46a, 20w51a, 21w03a, 1.16.5, 21w05b, 21w06a, 21w07a, 21w13a, 1.17, 1.17.1, 1.18, 1.18.1, 1.18.2, 1.19 Pre-release 1, 1.19,

1.19.1 Release Candidate 2, 1.19.2, 1.19.3, 1.20.1, 1.20.2, 23w42a, 23w43b, 1.20.3 Pre-Release 1, 1.20.4, 23w51b, 24w13a, 1.21 Pre-Release 2 (includes snapshots)

ORIGINAL DESCRIPTION

The bug

When a mob transforms into another mob (ex. zombie villager → villager, mooshroom → cow), it loses most of its NBT tags. This also affects conversions to the same mob, like slimes and magma cubes splitting.

It is worth noting CustomName and NoAI are preserved (

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), so other tags are expected to be saved as well.

Examples

The following is a list of tags that aren't transferred between entities.

From

Husk

Zombie

Skeleton

Zombie Villager

Villager

Villager

Pig

Piglin

Piglin Brute

Hoglin

Slime

Magma Cube

Tadpole

Mooshroom

To

Zombie

Drowned

Stray

Villager

Zombie Villager

Witch

Zombified Piglin

Zombified Piglin

Zombified Piglin

Zoglin

Smaller Slime

Smaller Magma Cube

Frog

Cow

8
ArmorDropChances
Attributes
CanBreakDoors
N/A
N/A
N/A
N/A
N/A
N/A
N/A
N/A
N/A
N/A
N/A
CanPickUpLoot
9
CustomName
20
20
CustomNameVisible
DeathLootTable
DeathLootTableSeed
FallFlying
8
8
8
8
N/A19
N/A19
8
8
Fire
N/A5
N/A15
N/A15
N/A15
N/A15
N/A15
N/A23
ForcedAge
N/A
N/A
N/A
N/A

N/A

N/A

N/A

N/A

N/A

N/A

N/A

N/A

N/A

Gossips

N/A

N/A

N/A

N/A

N/A

N/A

N/A

N/A

N/A

N/A

N/A

N/A

HandItems

8

8

8

8

N/A19

N/A19

8

8

HandDropChances

Health

N/A10

N/A21

N/A21

HurtByTimestamp

HurtTime

)

Invulnerable

N/A10

Leash

LeftHanded

Motion

NoAI

N/A4
N/A4
N/A6
11
13
13
N/A17
N/A17
N/A18
NoGravity
Offers
N/A
N/A
N/A
N/A
N/A
N/A
N/A
N/A
N/A
N/A
N/A
N/A
N/A
N/A
OnGround
Passengers1
N/A
N/A
PersistenceRequired
14
16
16
PortalCooldown
RootVehicle2
12
Rotation3
22
22
SleepingX, SleepingY, SleepingZ
Silent
Tags
UUID
23
23
VillagerData
N/A

N/A
N/A
N/A
N/A
N/A
N/A
N/A
N/A
N/A
N/A
N/A
Xp
N/A
N/A
N/A
N/A
N/A
N/A
N/A
N/A
N/A
N/A
N/A
N/A
N/A
N/A
wasOnGrou

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-88967>